

555 Timer PCB in Astable configuration

1. OBJECTIVES

- To understand and apply how to build a PCB from conceptualization to finished product including component selection, schematic drafting, PCB layout and manual assembly process.
- To drive multiple LED's with a 555 Timer output and measure the Rth resistance, noise on 5V rail and functionality.

2. COMPONENTS USED/BOM

1. NE555DR
2. Resistor – 47ohm, 300ohm, 1kohm, 10kohm.
3. LED – 5 nos
4. Power plug
5. Header pins – 2 nos
6. Capacitor – 1uF , 22uF x2

3. PLAN OF RECORD (POR)

The PCB circuit should contain the following

- A “AC to 5VDC” Power plug for powering the circuit.
- LED indication to know if power supply is ON
- Header pin to isolate 555 timer section from power supply section
- 555 timer to provide 5Vp-p square wave signal with around 500Hz and 66% Duty cycle.
- Header pin to isolate LED section from 555 timer section
- 4 LED's in parallel to be driven by the 555 timer output signal
- 4 different value current limiting resistors for LED's namely 47ohm, 300ohm, 1kohm, 10kohm.
- Test points to measure 5V power rail, 555 timer output and current through four different resistors.

4. ROUGH SKETCH AND CIRCUIT IMAGE

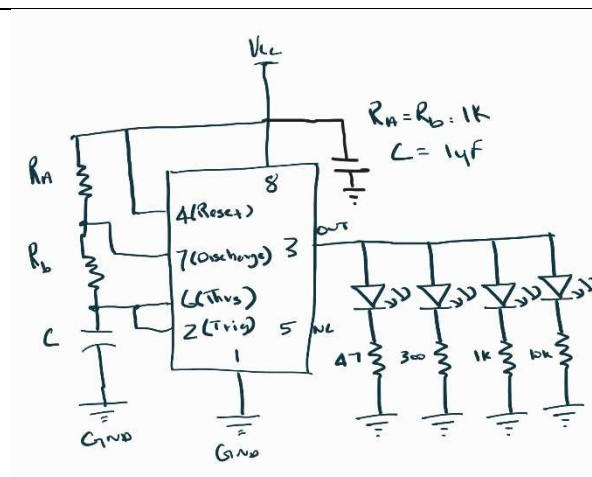


Fig 1 - Rough Sketch

5. ESTIMATION OF COMPONENT VALUES

The 555 timer need to drive 4 parallel LED's and should have enough sourcing capabilities. To estimate the current through the LED, consider the approximate voltage drop across the LED's as 2V which is common. The signal is 5Vp-p square wave.

So,

- The current through the 47ohm = $\frac{5-2}{47} = 63.82mA$ (estimation by ohms law)
- The current through the 300ohm = $\frac{5-2}{300} = 10mA$ (estimation by ohms law)
- The current through the 1kohm = $\frac{5-2}{1000} = 3mA$ estimation by ohms law)
- The current through the 10kohm = $\frac{5-2}{10000} = 0.3mA$ (estimation by ohms law)

The total current sourced = 77.12mA

NE555DR sourcing and sinking capabilities are up to 200mA which satisfies our requirements.

• FREQUENCY

$$\text{Frequency (f)} = \frac{1.44}{(Ra+2R) * C} \quad (\text{as specified in the NE555 datasheet by Texas Instruments})$$

I considered **Ra & Rb = 1Kohms** and **C = 1uF** to get desired output frequency close to 500Hz

$$\text{Substituting above values we get, } f = \frac{1.44}{(1000+200) * 10^{-6}} = 480Hz \quad (\text{which is close to our desired frequency})$$

• DUTY CYCLE

$$\text{On time Duty cycle } Ton = \frac{tH}{tH+t} = 1 - \left(\frac{Rb}{Ra+2R} \right) = 1 - \left(\frac{1000}{3000} \right) = 66.6\%$$

$$\text{Off time Duty cycle } Toff = \frac{Rb}{Ra+2R} = \frac{1000}{3000} = 33.3\%$$

(as specified in the NE555 datasheet by Texas Instruments)

$$f = \frac{1}{T} \cong \frac{1.44}{(RA + 2RB) \times C}$$

$$\text{Output driver duty cycle} = \frac{tL}{T} \cong \frac{RB}{RA + 2RB}$$

$$\text{Output waveform duty cycle} = \frac{tH}{T} \cong 1 - \frac{RB}{RA + 2RB} = \frac{RA + RB}{RA + 2RB}$$

Figure 2

6. SCHEMATIC DIAGRAM

555-TIMER-ASTABLE-CONFIGURATION

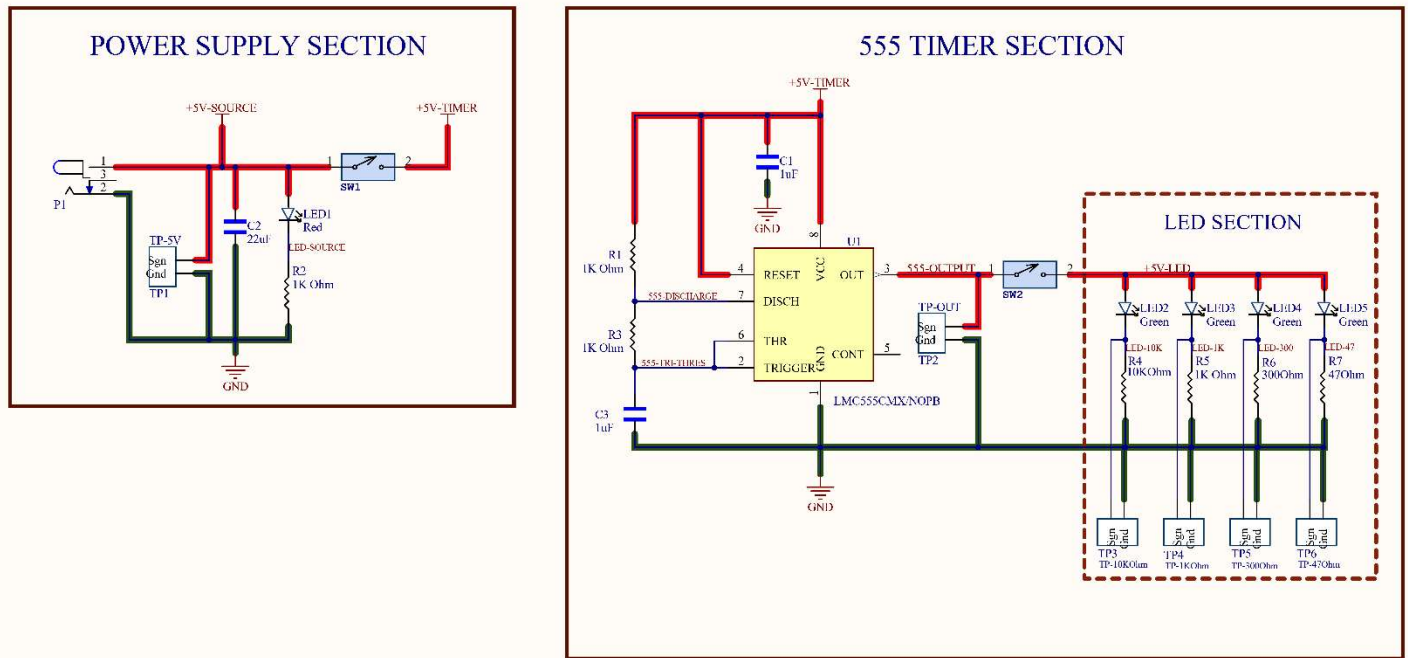


Figure 3 - Schematic diagram in Altium software

7. PCB LAYOUT

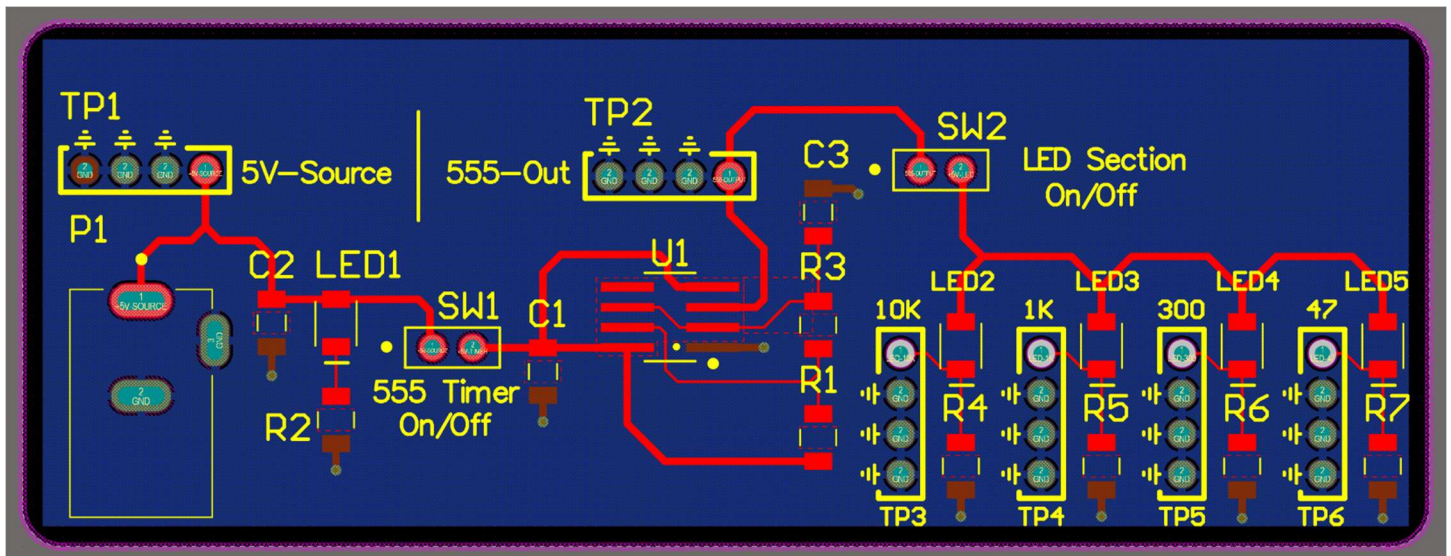


Figure 4 - PCB Layout in Altium software

8. BEFORE AND AFTER MANUAL ASSEMBLY

• BEFORE MANUAL ASSEMBLY

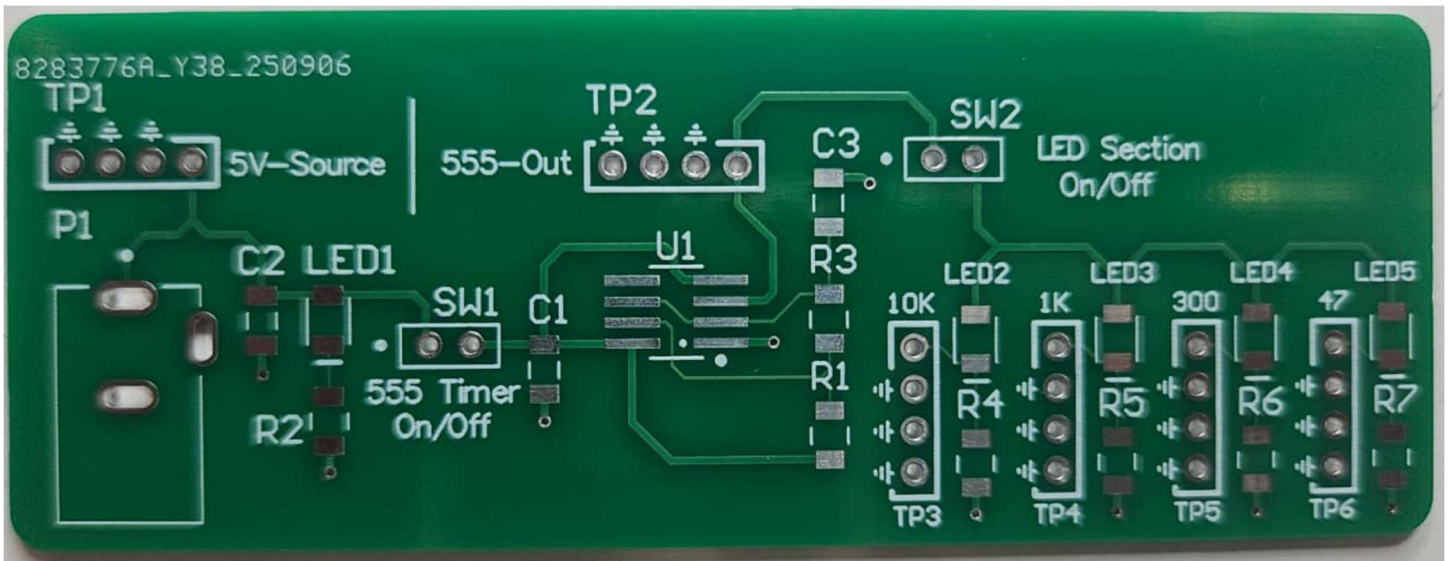


Figure 5 – Before Assembly

• AFTER MANUAL ASSEMBLY

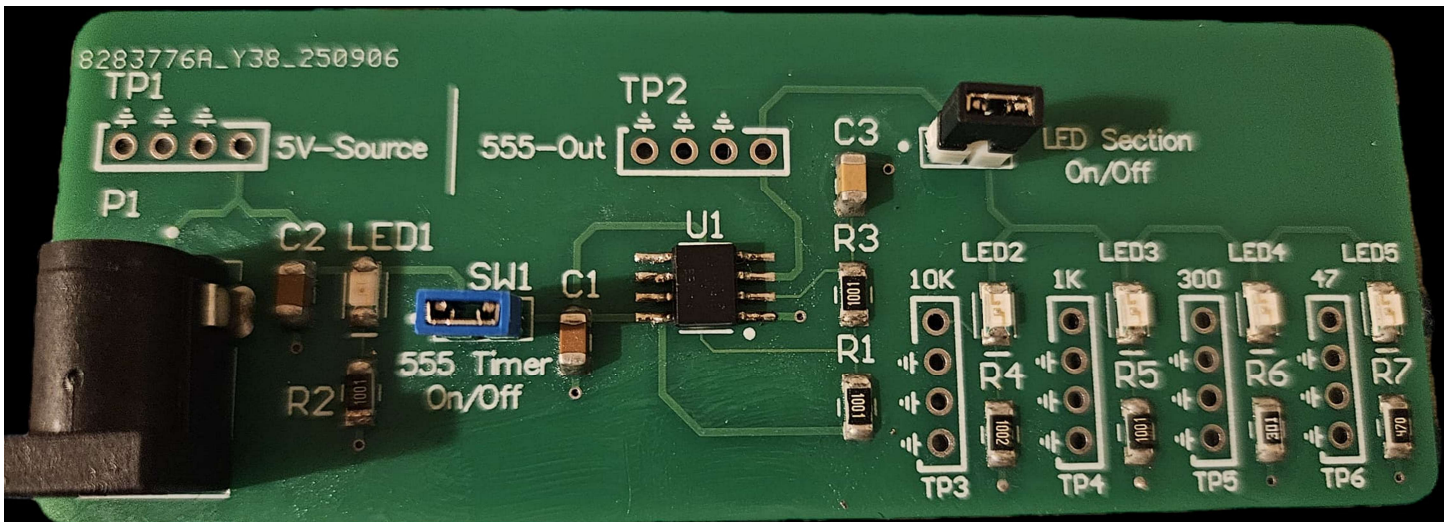


Figure 6 - After assembly

9. WORKING CRITERIA

- 5V indicator LED should turn on when powered up and waveform should be measured in TP1
- With SW1 on, 555 timer should give a 500Hz square wave with 66% duty cycle output measured in TP2
- With SW2 on, LED's should light up with different intensities based on load current. Current should be measured across each resistor using testpoints TP3, TP4, TP5, TP6 for 10Kohms, 1Kohms, 300Ohms, 470Ohms respectively.

10. AFTER POWERING UP

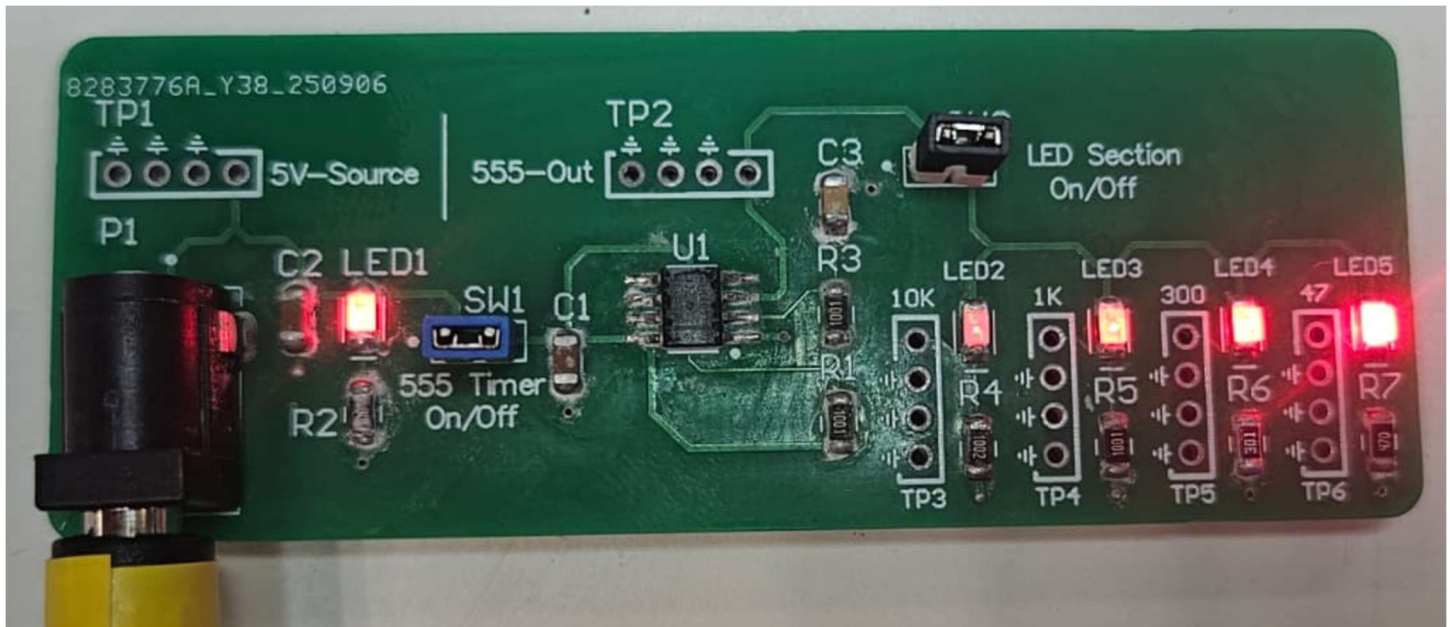


Figure 7 - Board after powering up

- The 5V indication LED lit up showing there was no issues with the power supply. Measurements were also taken using TP1 to make sure of the same.
- After SW1 was turned on, output from 555 was measured using TP2 and a 5Vp-p square wave with 500Hz and 66% duty cycle was observed.
- After SW2 was turned on, the luminosity of LED's was observed and respective load currents were measured using TP3, TP4, TP5 and TP6.

11. DEBUGGING

The board turned on with no issues and all waveform generated had no issues and gave expected results. Hence no debugging was required on this board.

12. CALCULATING THE THEVENIN'S RESISTANCE

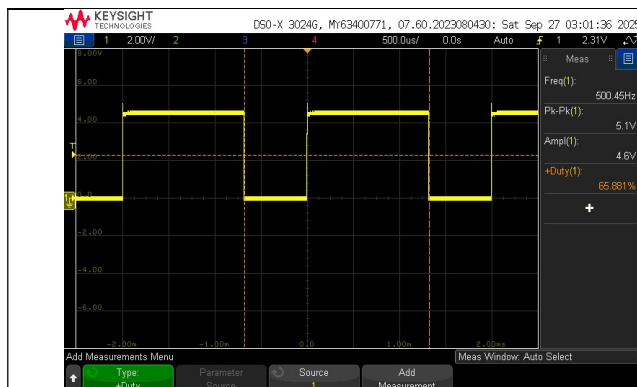


Figure 8 - 555 Timer output without Load

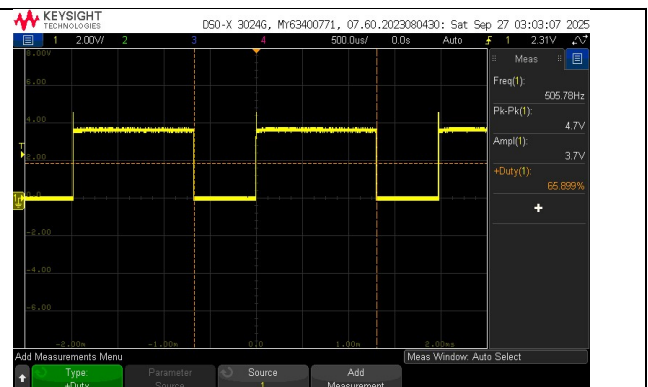


Figure 9 - 555 Timer output with Load

The above channel(yellow) shows the 555 Timer output measure at TP2 with and without Load.

- The amplitude of the output without load = 4.6V (open circuit voltage – V_{th})
- The amplitude of the output with load = 3.7V (Load voltage – V_L)
- The resistance seen by the 4 resistors = $\frac{1}{\left(\frac{1}{47}\right) + \left(\frac{1}{300}\right) + \left(\frac{1}{1000}\right) + \left(\frac{1}{10000}\right)} = 38.895 \text{ ohms } (R_{load})$

$$R_{th} = R_{load} \left(\frac{V_{th} - V_L}{V_L} \right) = 38.895 \left(\frac{4.6 - 3.7}{3.7} \right) = 9.461 \text{ ohms}$$

This shows that the 555 timer has an internal resistance of around **9.5 ohms** and whenever current is sourced a voltage drop occurs across this internal resistance making the 555 timer not an ideal source.

13. SWITCHING NOISE ON 5V POWER RAIL

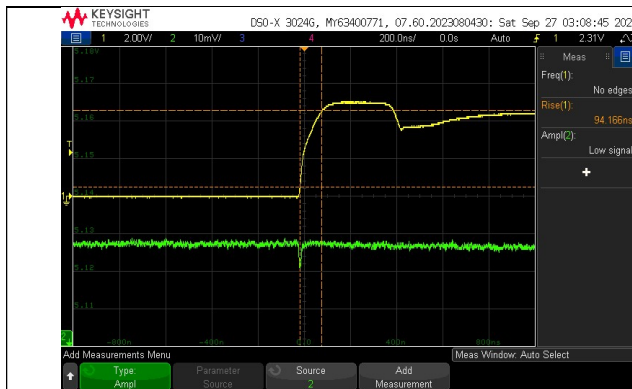


Figure 10 - 555 timer output rise time without load

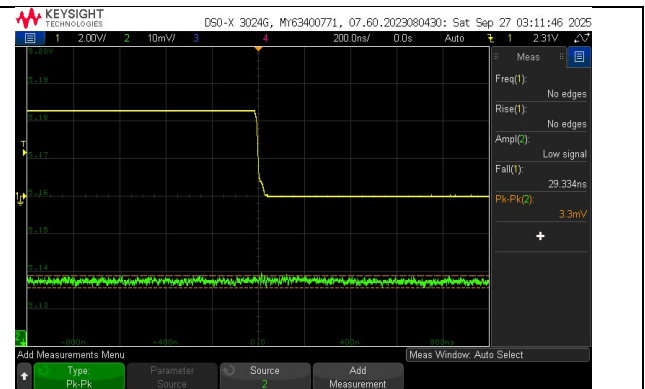
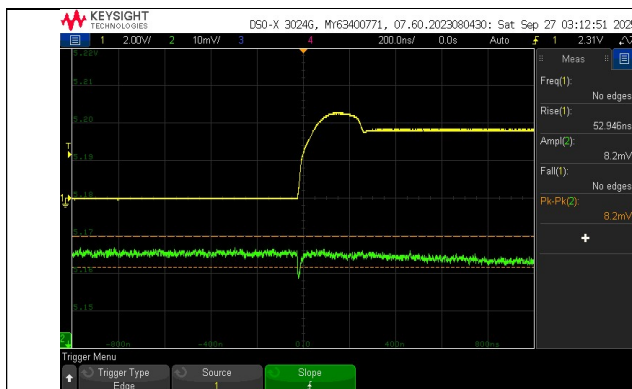
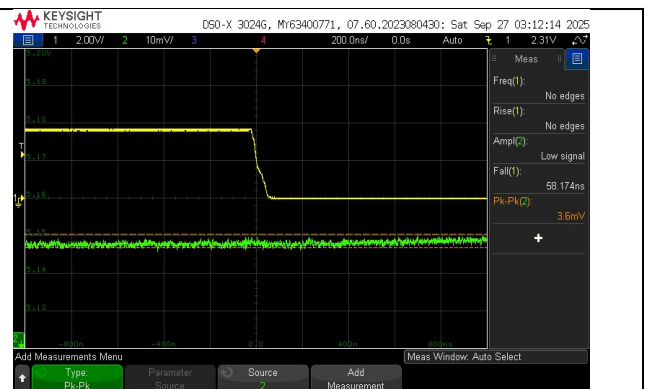


Figure 11 - 555 timer output fall time without load

In the above Figures, Channel 1(yellow) shows the 555 timer output and Channel 2 (green) show the 5V power rail. The rise time measured is 94.166ns and fall time measured is 29.334 ns. The noise on the voltage rail is around 8mV during rise time and 3.3mV during fall time, both of which are not significant value for our application. The voltage noise occurs due to voltage drop in the loop inductance of the power rail in the di/dt region.



555 timer output rise time with load

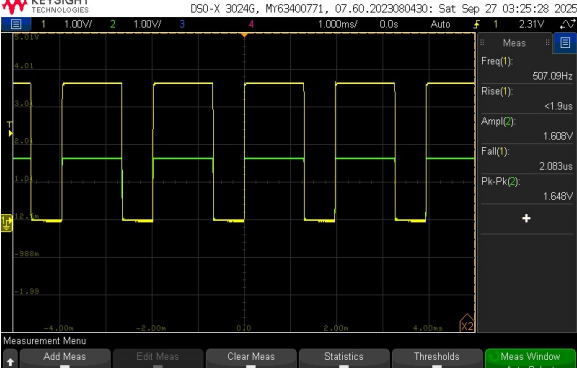
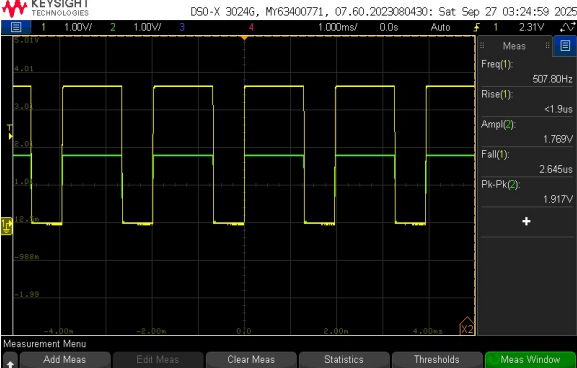
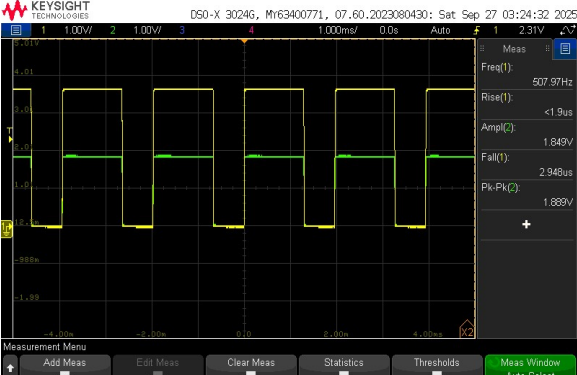
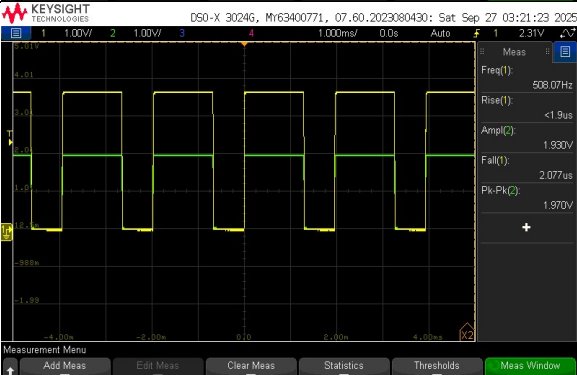


555 timer output fall time with load

In the above Figures, Channel 1(yellow) shows the 555 timer output and Channel 2 (green) shows the 5V power rail. The rise time measured is 52.946ns with 8.2mV voltage noise in the power rail and the fall time measured is 58.174ns with 3.5mV voltage noise in the power rail. There is not much difference in the voltage noise between no load condition and load condition even after the reduction in rise time because of the decoupling capacitor placed near the 555 Timer.

14. CURRENT CALCULATIONS FOR LOAD RESISTORS

In the Below figures, Channel 1(yellow) shows the 555 timer output and Channel 2 (green) show the Voltage across the resistor. Current is then calculated using ohms law. By comparison, 1kohm comparatively has better luminosity and takes less current which makes it a good option for lighting the LED's.

	Voltage across 47 ohms = 1.6V $\text{Current} = \frac{1.6}{47} = 34mA$
	Voltage across 300 ohms = 1.77V $\text{Current} = \frac{1.77}{300} = 5.9mA$
	Voltage across 1K ohms = 1.85V $\text{Current} = \frac{1.85}{1000} = 1.85mA$
	Voltage across 10K ohms = 1.93 $\text{Current} = \frac{1.93}{10000} = 0.193mA$

15. BEST DESIGN PRACTICES FOLLOWED

1. Decoupling capacitor is placed near the IC to minimize voltage noise due to the voltage drop across the loop inductance in the power rail.
2. Proper labelling was given to all switches and test points for better understanding of the board
3. Switches were placed between different sections and proper indicating LEDs were placed to aid for debugging
4. Continuous ground plane was provided on bottom layer for proper return path and thicker trace widths were used for power nets.

16. IMPROVEMENTS TO BE IMPLEMENTED

More soldering experience to be gained to get better solder joints.

17. CONCLUSIONS

The 555 Timer in Astable mode met all goals and worked on first power-up. The 555 timer output had a 5Vp-p, 500Hz and 66% duty signal as calculated. The internal Thevenin's resistance was measured at 9.5 ohm. Power-rail noise stayed low due to proper decoupling capacitor placement. Good layout practices were carried out and manual assembly was done with no error.